

Detection and Prediction of Landslide Vulnerability through Case Study using DInSAR Technique and U-net Model

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Abstract—Landslides are among the deadliest natural catastrophes because they usually occur after other severe disasters, causing enormous damage to already vulnerable systems. In recent years remote sensing methodologies have grown prominence in landslide monitoring and prediction due to their ability to mitigate hazards. In remote sensing, the Interferometry technique is employed in Synthetic Aperture Radar. In this methodology, the DInSAR technique is used for the phase difference calculation over a specific area using the sensors of the Sentinel satellites. SR-530 landslide is the chosen field of study for assessing the properties and parameters required to detect and predict landslides. The Digital Elevation model can be generated from the Sentinel satellite imagery using the DInSAR technique. Apart from DEM, Slope and Normalized Difference Vegetation Index are also used to detect and quantify the presence of land deformation and erosion. Natural color (RGB) image is also considered in landslide detection. These parameters are used to predict the activity of landslides using the U-net Model through training on various data samples across the globe. Immediate recognition can be made once the landslide starts showing its signature like land deformations, changes in erosion, vegetation, etc. From the obtained results, forthcoming landslides can be detected.

Keywords—Landslide detection, Differential Interferometric Synthetic Aperture Radar technique, U-net Model, Normalized Difference Vegetation Index, Remote Sensing, Digital Elevation Model, Sentinel Satellite.

I. INTRODUCTION

In the last ten years, earthquakes and heavy rains-induced landslides, and slope failures led to two billion USD in economic loss and around 10,000 deaths. With its ability to observe in broad, cloud-free swaths both day and night, Currently, Synthetic Aperture Radar (SAR) is essential for timely landslide monitoring. In particular, landslide risk assessment, ongoing monitoring, and early detection techniques commonly make use of interferometric SAR (InSAR). To precisely pinpoint landslide regions, supplementary data obtained from a DEM (digital elevation model), such as slope and curvature, were also utilized [1]. On March 22, 2014, landslides tragically caused the loss of life and devastation of property in northwest Washington. About 40 homes and other buildings, as well as about a mile of State Route 530, were covered in landslide material. 43 people died as a result of it at Steelhead Haven, a town close

to Oso, Washington. The adverse effects of the landslide in the Oso region are mentioned in [2]. The Stillaguamish River gradually eroded back to close to its pre-landslide elevation in the 6 to 8 weeks that followed the landslide, emptying the remaining extra water by the middle of May. The effect of debris on the river is also mentioned in [2]. Figure 1 represents the area of debris flow in the Oso region [3].



Fig 1. Oso debris flow region [3]

In this paper, the notion of supervised classification utilizing the U-net Model is proposed as a way to identify landslides by integrating parameters like the area's DEM, Slope, NDVI, and RGB Images.

II. RELATED WORK

In [4] the authors proposed a system for landslide detection using the pixel-based deep learning approach from the sentinel-2 imagery. It utilizes ResU-net, Object-based image analysis, and ResU-net-Object-based image analysis. The area under study is the Taiwan region after the Morakot typhoon in august 2013. The key drawback is that only a three percentage-point improvement in recall value is obtained using the combined strategy, which is less significant than the improvement in precision.

According to [5], the authors proposed a system for “Rainfall-Triggered Landslides in Crete (Greece) Based on Spatial Detection and Hazard Mapping”. It deals with the detection of landslides caused due to heavy rainfall in the Mediterranean Crete region. The main disadvantage is that the conditioning factors and the threshold values used in it can slightly mismatch with the location of the landslide.

As per [6], the authors proposed a system to detect landslides through densely connected convolution networks.

It takes advantage of Dense Nets and its modified techniques which help in detecting landslides. The main disadvantage is that it is unable to detect micro landslides.

According to [7] authors proposed a system to detect landslides in the Cyprus region. This research intends to show how the use of publicly available and open-access i.e., Copernicus sci-hub datasets. It also showed the methodology, and how to interact with the European Space Agency's SNAP (Sentinel's Application Platform) open-source processing software.

In [8], the authors proposed work that analyses several InSAR-related landslide study applications in Italy. In this study, more than 250 papers were examined. The purpose of this research is to demonstrate how satellite interferometry may be used to monitor and map landslides in Italy.

According to [9], the authors proposed a system to detect landslides in 3 chosen areas (Etna, California Highway 1, Anargyroi Greece) which have unique factors that cause the landslide. Sentinel-1, a C-band SAR satellite, was used to collect the data for this investigation. The main disadvantage is that it is affected by layover and shadowing principles.

In [10], the authors proposed a system which uses to evaluate the potential of machine learning techniques and various deep-learning convolution neural networks for landslide detection, they used optical data from the Rapid Eye satellite and topographic parameters. The case study region is located in Nepal's Rasuwa district's southern region. The biggest drawback is that the result's general accuracy was significantly decreased by topographical information.

According to [11], authors proposed a system to detect the landslides in mountain regions which is difficult because of significant decorrelation. L-band SAR data from the ALOS-2 PALSAR-2 satellite which is fully polarimetric are used in this investigation. The main disadvantage is that the model mistakenly detects the forest area as a landslide due to the long temporal distance.

III. TESTSITE AND DATASET

A. Test Site

The Oso Landslide was more catastrophic because it was bigger, traveled farther, and had more force than the others. The adverse effects of the Oso landslide are mentioned in [2]. Many studies and research have been conducted on this landslide which helped us in studying and analyzing the landslide-causing factors. Table 1 lists the properties of the sentinel satellite at the time of considering the data. It includes the pre-event and post-event data dates for the considered site from the Copernicus open-access hub.

Table 1: Sentinel-1 InSAR Level 2 NISAR Format Data used for detecting the landslides in this study

Properties	Value
Observation dates: Post-Event - Pre-Event	2015-03-01 - 2014-03-01
Temporal Baseline	12 days
Orbit, Look direction	Ascending, right
Azimuth angle	-9.7°
Incidence angle	43.9°
Polarization	VH
Phase Identifier	1

B. Dataset

The data about various landslide areas are fetched from the landslide4sense website. It provides the data for a total of 4844 locations across the globe. The dataset includes the Train, Test, and Valid data separately. The train folder consists of 3799 samples, the test folder consists of 245 samples and the validation folder consists of 800 samples. The 'Train' data can be used for training purposes and 'Test' data can be used for test purposes. So, there is no need for splitting the data as the data is provided directly. The data consists of sentinel-2 satellite B1-B12 multispectral bands. Along with these, the dataset also contains DEM and the Slope of the corresponding location. The data provided will be of approximately 10m resolution. The images will be of size 128*128 pixels. The train, test, and valid directories contain the .h5 files. In these files, all the data corresponding to the location will be stored [12]. Large amounts of data are stored in the form of multi-dimensional arrays using hierarchical data formats (HDF). The format is primarily used to hold well-organized scientific data that can be quickly retrieved and analyzed.

IV. METHODOLOGY

A. Field Study

The field study involves determining the parameters required for detecting landslide susceptibility. The criteria are that, when these determined parameters are considered as the primary parameters for landslide detection, the majority of the landslides are to be detected. Upon analyzing various landslides, SR-530 is found to be the best landslide as it is the most devastating landslide in the USA. Figure 2 represents the Oso landslide in-situ image [14]. The Oso landslide simulations are in [13].



Fig 2. Oso Landslide in-situ image [14]

Upon studying the SR-530 landslide from [2] we finalized the following parameters for landslide susceptibility detection:

- Digital elevation model (DEM)
- Slope
- Normalized Difference Vegetation Index (NDVI)
- Natural Color Image (RGB)

There are several ways to obtain DEM. The sudden change in elevation is the major cause of landslides [15]. When there is a change in land topography (when a landslide starts showing its signature), there will be a change in DEM. By identifying this change, the occurrence of landslides can be predicted.

The Slope is generated from the DEM. It denotes the steepness at each cell in the DEM raster. The change in Slope in a region can be used to detect landslides because it determines the occurrence and intensity of the landslide [16].

The Normalized Difference Vegetation Index is a representation that quantifies the vegetation of a region. Landslides that have recently occurred are frequently accompanied by the occurrence of lodged or covered vegetation and exposed bedrock, or bare soil. As a result, we may see clear differences between the landslide region and the surrounding region in terms of vegetation if a landslide occurs in a vegetated area [17]. By identifying this change, we can easily detect landslides.

The RGB Image obtained from the open-access Copernicus sci-hub and Alaska satellite facility can be used for landslide detection. Using the RGB Image as a parameter, the model will be able to detect the landslide by analyzing the patterns of landslides. [18]. By training on the RGB images of landslide areas and non-landslide areas, the occurrence of any new landslide can be easily identified.

B. Dataset Collection and Pre-processing

As all our required conditions are satisfied in the dataset provided by the landslide4sense challenge, we opted for the dataset from it. The downloaded dataset contains majorly three directories namely “Train”, “Test” and “Valid”. The 3799 samples in “Train”, 245 samples in “Test” and 800 samples in “Valid” directories will be used for training, testing and validating the model respectively. This gives the advantage of avoiding the necessity to split the data. All the data corresponding to a single geographical location are combined and stored as a .h5 file [19]. The data in a .h5 file will be read and stored in a Numpy array. This array’s shape will be (128,128,14). The (128,128) indicates the size of each image and 14 indicates the number of images present in it. RGB images can be obtained by extracting the images in 2,3,4 dimensions. The 2nd, 3rd, and 4th dimensions contain blue, green, and red bands data. Combining all these bands’ information, we can obtain an RGB image. Similarly, by extracting the image in the 13th dimension, we can obtain DEM. If we extract the image at the 14th dimension, we obtain the Slope. After extracting the images at the 8th and 4th dimensions and applying the formula (5), we obtain the NDVI. The process of extracting the data from the .h5 file is mentioned in [20]. In this way, the required data for training the model is extracted from the .h5 file. Table 2 describes the details of the dataset after pre-processing.

Table 2: Dataset details

Tag	Dimensions
Train Input Data size	(3799,128,128,6)
Train Output Data size	(3799,128,128,1)
Test Input Data Size	(800,128,128,6)
Test Output Data Size	(800,128,128,1)
Valid Input data size	(245,128,128,6)
Valid Output data size	(245,128,128,6)
Image Dimensions	(128,128)
Images resolution	10m

The first value in each value of the dimension column indicates the number of samples of data. The second and third numbers indicate the image dimensions. The fourth number indicates the number of parameters or images present in each sample. In six, the first three dimensions will be used to store the RGB data, the fourth dimension for storing the NDVI, fifth dimension for storing the RGB image data of a geographical location. In the case of output data, the one indicates the label of the location

C. Model Training

Module 3 deals with training the dataset using the U-net Model. The generic convolutional neural network concentrates on classifying images when its inputs are set as parameters, and its output is a single label. But the U-net model helps in not only classifying but also pinpointing the occurrence. In the case of landslide detection too, detecting the occurrence of landslides along with the detection of location is required. So, the U-net model is chosen. The trained data is given as input to the model [21].

In the contractive path, the conv2D layer is added with the required number of units and the appropriate activation function used. Here the “relu” activation is used [22]. The layers are initialized with proper units and activation function as “relu”. We opted for the “relu” because of its advantages mentioned in [23]. The characteristics of the “relu” activation function are mentioned in [23] and [39]. Four levels connecting with conv2D layers are created with the activation function as “relu”. The model is also associated with the usage of Modelcheckpoint() to prevent overfitting [26]. The model is compiled using the “adam” optimizer [24] and metrics such as ‘accuracy’, ‘recall’, ‘f1’, and ‘precision’ [25]. The confusion matrix consists of four major terms: True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). These terms are explained in [40]. The Accuracy, precision, recall, f1-score values are obtained by applying the formulae (1), (2), (3), (4) respectively on confusion matrix terms namely TP, TN, FP and FN [40].

$$\text{Accuracy} = \frac{TN+TP}{TN+TP+FN+FP} \quad (1)$$

$$\text{Precision} = \frac{TP}{TP+FP} \quad (2)$$

$$\text{Recall} = \frac{TP}{TP+FN} \quad (3)$$

$$\text{F1-Score} = 2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

Table 3 denotes the details of the training.

Table 3: Training Details

Tag	Value
Model input shape	(128,128,6)
epochs	100, 16
Batch Size	16
Callbacks	ModelCheckPoint
Callbacks monitor value	val_f1_m
Parameters	Accuracy, precision, recall, f1

D. Generation of Parameters

This module is about the generation of DEM, NDVI, Slope, and RGB images for the area of study in order to validate the trained model.

Dem Generation

Using the stereoscopic effect brought on by minute variations in observation location between the two images, interferograms can be utilized to create digital elevation models (DEMs). The methodology for the generation of DEM using the DInSAR technique is mentioned in [27]. DEM can be generated using the SNAP tool. For this, it is necessary to take into account the sentinel satellite data at two separate periods. The sentinel-1 data can be downloaded from the Copernicus open-access hub. Table 4 indicates the details of the considered Input products and their processing parameters.

Table 4: Description of Input products

Input product	Subswath	Polarization	Bursts	Output name
S1A_IW_GRDH_1SDV_20220805_....	IW	VH	3 – 5	S1A_20220805_sp lit
S1A_IW_GRDH_1SDV_20220915_....	IW	VH	1 – 3	S1A_20220915_sp lit

The two products considered must vary within the shorter time variation periods. This helps in analyzing the land deformations. DEM can be generated using the SNAP tool [28]. Figure 3 represents the graph builder [29]. It represents the following process for DEM generation

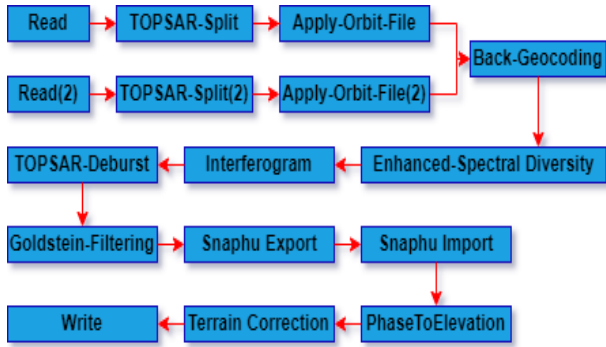


Fig 3. DEM generation process

Slope Generation

Slope is obtained from the DEM. The slope can be generated by using the “Slope” tool in ArcGIS where the input raster will be the DEM [30].

NDVI Generation

The normalized difference vegetation index gives the change in vegetation index at a given location. Based on the satellite, the methodology for generating the NDVI will be changed. For the sentinel-1 satellite imagery, the NDVI generation formula is as follows [31].

$$NDVI = \frac{NIR + Red}{NIR - Red} \quad (5)$$

By using the “Raster Calculator” tool in ArcGIS, we can obtain the NDVI for our required location from NIR and

Red bands. NDVI is obtained by applying equation (5) on each and every pixel of NIR and Red bands.

RGB Image Generation

The sentinel-1 satellite also provides the natural color image. This RGB image can be downloaded from the scihub. The obtained RGB image has to be clipped with the help of the shape file of the SR-530 region using the “extract by mask” tool in ArcGIS [32]. This clipped image will contain three dimensions namely red, blue, and green. To store RGB images, we have to allocate a numpy array with shape (128,128,3) where 128,128 indicates the size of image and 3 indicates the dimensions.

Figure 4 indicates the process flow followed. It represents the sequence of steps followed in this study.

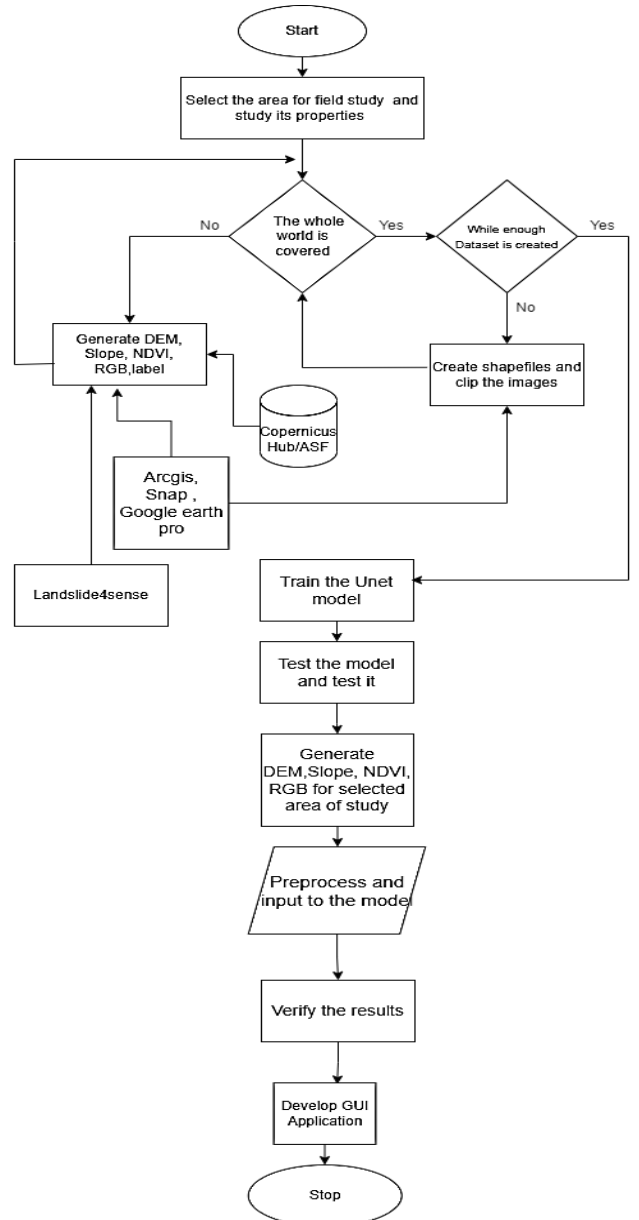


Fig 4. Process flow

E. Model Evaluation

The module-5 is about validating the trained model. In order to check the trained model apart from the validation

data provided in the dataset, we tested the model with the manually generated DEM, NDVI, Slope, and RGB data of the SR-530 landslide. The verification will be done in such a way that the label generated by the model shows the landslide in the same region where the actual landslide occurred. This can be done by providing the SR-530 data to the model as input and predicting it using the predict() in the keras library [33]

V. ALGORITHMS

A. Dataset Collection and Pre-processing

Input: Data collected from landslide4sense challenge.

Output: Data for training, testing, and validation.

1. Traverse through the Train, Test, and Valid Directories in the dataset.
2. k = number of files in the directory
3. Initialize $val = np.zeros((k, 128, 128, 6))$
4. Initialize $label = np.zeros((k, 128, 128, 1))$
5. For $i = 0$ to $i = k$ and increment by 1
6. $val[i, :, :, 0:3] = \text{RGB data}$
7. $val[i, :, :, 3] = \text{NDVI data}$
8. $val[i, :, :, 4] = \text{Slope data}$
9. $val[i, :, :, 5] = \text{Elevation data}$
10. $label[i, :, :, 0] = \text{Label data}$

B. Model Training

Input: NumPy arrays generated from the dataset

Output: Trained model.

1. Define recall, precision, and f1 functions.
2. input shape = (128,128,6)
3. Use “relu” activation function and create layers in the contraction path following [22]
4. Create layers in the expansion path following [34]
5. Compile the model with an “adam” optimizer and metrics such as accuracy, f1, precision, and recall. Use ModelCheckpoint in the callbacks library for early stopping [25]
6. Fit the model with train data and start training [35]
7. Evaluate the model using evaluate() in the keras module test data [33].
8. Obtain the accuracy and loss of the model.

C. Model Evaluation

Input: SR-530 Landslide data

Output: Landslide susceptibility of the SR-530 landslide

1. Initialize $val = np.zeros((1, 128, 128, 6))$.
2. Read each image using imread() in the “image” library [36] and store it into different arrays with names rgb, ndvi, slope, and dem.
3. Resize the image to 128*128 size.
4. $val[0, :, :, 0:2] = \text{rgb}$
5. $val[0, :, :, 3] = \text{ndvi}$
6. $val[0, :, :, 4] = \text{slope}$
7. $val[0, :, :, 5] = \text{dem}$
8. $\text{pred_result} = \text{predict}(val)$ [33]
9. Plot the pred_result array and visualize

VI. RESULTS AND ANALYSIS

A. Software Specifications and Tools

In this study, the ArcGIS tool is used to extract the Oso region’s data and also used to obtain the Slope and NDVI.

ArcGIS is geospatial software used for viewing and analysing geographic data. ArcGIS performs pixel-level modifications on rasters. The “Slope” tool in ArcGIS helps in calculating the slope from DEM. The “raster calculator” helps in performing any manipulations on rasters at pixel level. This helps in NDVI generation. “Extract By Mask” tool can be used to extract the required part from a raster using shape file.

SNAP tool is also used in this study to generate the DEM. It is the application platform for the Sentinel satellite. It is utilised for modelling, visualisation, and image processing [41]. The SNAP tool’s unwrapping tool aids in the recovery of clear phase data from a 2-D array of phase values. With the use of the Phase to Elevation tool, phase data can be transformed into elevation data.

B. Model Training

Training is done training with the U-net model. The metrics used for training include precision, recall, f1, and accuracy. Table 5 represents the model results

Table 5: Training results

Metric	Value
Loss	3.3%
Accuracy	95%
F1 score	71%
Precision	78%
Recall value	68%

Figure 5 represents the U-net model precision graph.

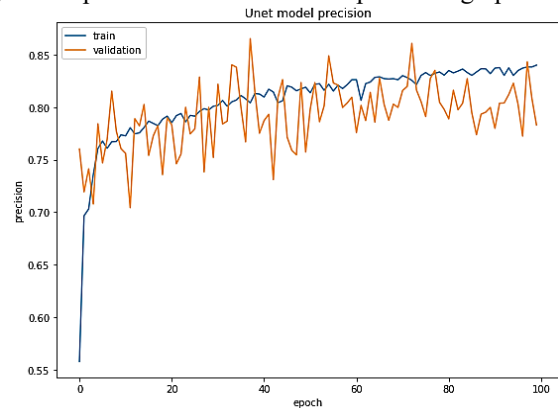


Fig 5. U-net model precision

Figure 6 represents the U-net model loss graph.

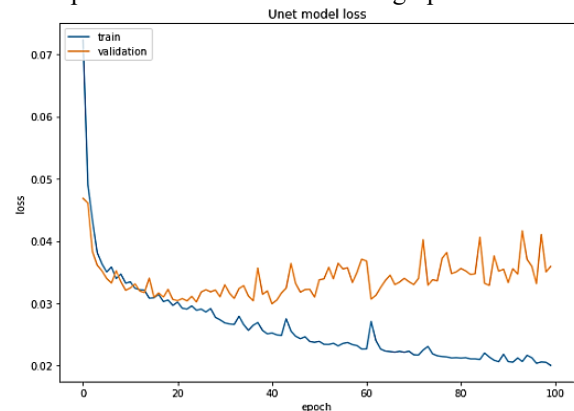


Fig 6. U-net model precision

C. Generation of Parameters

Figure 7 represents the phase data and coherence data after interferogram generation, the phase unwrapping result after the snaphu unwrapping process using the snaphu tool, and the elevation data after converting the phase data to elevation data (DEM). These are the results of the DEM generation process for data corresponding to a huge block of the area containing the Oso region.

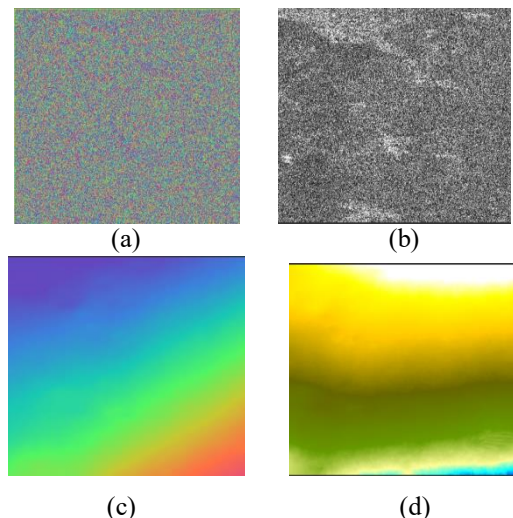


Fig 7(a). phase data 7(b). Coherence data 7(c). phase unwrapping result 7(d). Elevation data

Figure 8 represents the RGB, NDVI, Slope, and DEM images generated oso region and plotting them using the matplotlib [37]. Oso region DEM has to be extracted using the shape file of the Oso region from the 7(d). Shapefile generation is mentioned in [38].

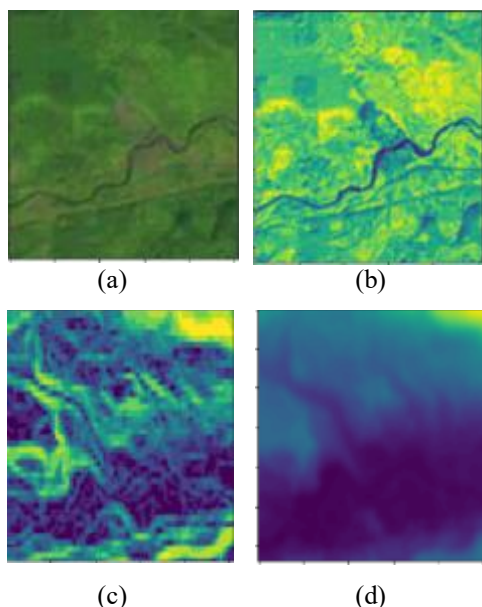


Fig 8(a). RGB 8(b). NDVI 8(c). Slope 8(d). DEM

D. Model Evaluation

Validation of the model is done using the manually generated data of the Oso landslide and also with the validation data provided in the challenge. Figure 9 represents the behaviour of the model when the data from the validation directory of the challenge is provided.

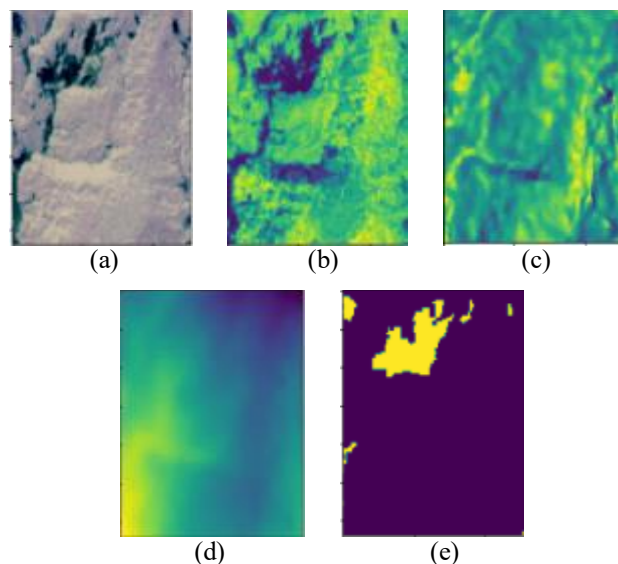


Fig 9(a). RGB 9(b). NDVI 9(c). Slope 9(d). DEM 9(e). Predicted Label

Figure 10 represents the prediction of the Oso landslide when the manually generated DEM, NDVI, Slope, and RGB images are provided as input for the model. The inputs are mentioned in Figure 8. The yellow-colored region represents the landslide susceptible region i.e the oso area.

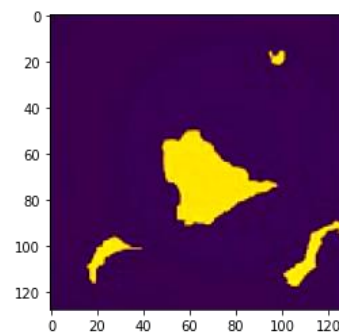


Fig 10. Oso landslide prediction

VII. CONCLUSION AND FUTURE WORK

A. Conclusion

This study involves the detection of landslides induced by land deformations. There are various causes for the landslide like rainfall, deforestation, human interventions, etc. Before any landslide occurs, some deformations can be observed on the Earth's topography. By analyzing the deformation patterns, we can predict the forthcoming landslide if it starts showing its signature. For this, continuous monitoring is required. If a landslide pattern is identified, immediately an alert can be made. We observed the DEM, Slope, NDVI, and RGB data of the location to find its landslide susceptibility.

B. Future Work

The future work includes finding the landslide susceptibility with both remote sensing and optical image analysis methods which includes the usage if drones revolving around the required area. The model can be made more accurate by using advanced deep-learning models. More parameters can be added apar from those used here.

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